MPG No. 3.1.6 2 [Type I, monoclinic] [G tensor]

* Rank 0 tensor.

$$[G]$$

$$G = \mathbb{G}_0^{(1)}$$

* Rank 1 tensor.

$$\begin{bmatrix} 0 & G_y & 0 \end{bmatrix}$$

$$G_y = \mathbb{G}_y^{(1)}$$

* Rank 2 tensor (s).

$$\begin{bmatrix} G_{xx} & 0 & G_{xz} \\ 0 & G_{yy} & 0 \\ G_{xz} & 0 & G_{zz} \end{bmatrix}$$

$$G_{xx} = \mathbb{G}_0^{(1)} - \mathbb{G}_u^{(1)} + \mathbb{G}_v^{(1)}$$
$$G_{xz} = \mathbb{G}_{xz}^{(1)}$$
$$G_{yy} = \mathbb{G}_0^{(1)} - \mathbb{G}_u^{(1)} - \mathbb{G}_v^{(1)}$$
$$G_{zz} = \mathbb{G}_0^{(1)} + 2\mathbb{G}_u^{(1)}$$

* Rank 2 tensor (a).

$$\begin{bmatrix} 0 & 0 & G_{xz} \\ 0 & 0 & 0 \\ -G_{xz} & 0 & 0 \end{bmatrix}$$

$$G_{xz} = -\mathbb{Q}_y^{(1)}$$

* Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & G_{xxy} & 0 \\ 0 & G_{yyy} & 0 \\ 0 & G_{zzy} & 0 \\ G_{yzx} & 0 & G_{zzy} \\ 0 & G_{yzx} & 0 \\ G_{xxy} & 0 & G_{yzx} \end{bmatrix}$$

$$G_{xxy} = \mathbb{G}_y^{(1)} - \mathbb{G}_y^{\alpha(1)} - \mathbb{G}_y^{\beta(1)}$$
$$G_{yyy} = 3\mathbb{G}_y^{(1)} + 2\mathbb{G}_y^{\alpha(1)}$$
$$G_{zzy} = \mathbb{G}_y^{(1)} - \mathbb{G}_y^{\alpha(1)} + \mathbb{G}_y^{\beta(1)}$$
$$G_{yzx} = \mathbb{G}_{xyz}^{(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & G_{xxy} & 0 \\ 0 & 0 & 0 \\ 0 & G_{zzy} & 0 \\ G_{yzx} & 0 & -\frac{G_{zzy}}{2} \\ 0 & G_{zxy} & 0 \\ -\frac{G_{xxy}}{2} & 0 & -G_{yzx} - G_{zxy} \end{bmatrix}$$

$$
\begin{aligned}
G_{xxy} &= 2\mathbb{G}_y^{(2)} - 2\mathbb{Q}_{xz}^{(1)} \\
G_{zzy} &= 2\mathbb{G}_y^{(2)} + 2\mathbb{Q}_{xz}^{(1)} \\
G_{yzx} &= 3\mathbb{Q}_u^{(1)} + \mathbb{Q}_v^{(1)} \\
\frac{G_{zzy}}{2} &= \mathbb{G}_y^{(2)} + \mathbb{Q}_{xz}^{(1)} \\
G_{zxy} &= -3\mathbb{Q}_u^{(1)} + \mathbb{Q}_v^{(1)} \\
\frac{G_{xxy}}{2} &= \mathbb{G}_y^{(2)} - \mathbb{Q}_{xz}^{(1)}
\end{aligned}
$$

* Rank 3 tensor (a).

$$
\begin{bmatrix}
G_{yzx} & 0 & G_{yzz} \\
0 & G_{zxy} & 0 \\
G_{xyx} & 0 & G_{xyz}
\end{bmatrix}
$$

$$
\begin{aligned}
G_{yzx} &= \mathbb{Q}_0^{(1)} - \mathbb{Q}_u^{(2)} + \mathbb{Q}_v^{(2)} \\
G_{yzz} &= -\mathbb{G}_y^{(3)} + \mathbb{Q}_{xz}^{(2)} \\
G_{zxy} &= \mathbb{Q}_0^{(1)} - \mathbb{Q}_u^{(2)} - \mathbb{Q}_v^{(2)} \\
G_{xyx} &= \mathbb{G}_y^{(3)} + \mathbb{Q}_{xz}^{(2)} \\
G_{xyz} &= \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_u^{(2)}
\end{aligned}
$$

* Rank 4 tensor (sss).

$$
\begin{bmatrix}
G_{xxxx} & G_{xxyy} & G_{xxzz} & 0 & G_{xxzx} & 0 \\
G_{xxyy} & G_{yyyy} & G_{yyzz} & 0 & G_{yyzx} & 0 \\
G_{xxzz} & G_{yyzz} & G_{zzzz} & 0 & G_{zzzx} & 0 \\
0 & 0 & 0 & G_{yzyz} & 0 & G_{yzxy} \\
G_{xxzx} & G_{yyzx} & G_{zzzx} & 0 & G_{zxzx} & 0 \\
0 & 0 & 0 & G_{yzxy} & 0 & G_{xyxy}
\end{bmatrix}
$$

$$
\begin{aligned}
G_{xxxx} &= \mathbb{G}_0^{(1)} + 2\mathbb{G}_0^{(2)} - \mathbb{G}_{4u}^{(1)} - \mathbb{G}_{4v}^{(1)} + 2\mathbb{G}_4^{(1)} - 2\mathbb{G}_u^{(1)} - 4\mathbb{G}_u^{(2)} + 2\mathbb{G}_v^{(1)} + 4\mathbb{G}_v^{(2)} \\
G_{xxyy} &= \mathbb{G}_0^{(1)} + 2\mathbb{G}_{4u}^{(1)} - \mathbb{G}_4^{(1)} - 2\mathbb{G}_u^{(1)} \\
G_{xxzz} &= \mathbb{G}_0^{(1)} - \mathbb{G}_{4u}^{(1)} + \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(1)} + \mathbb{G}_v^{(1)} \\
G_{xxzx} &= -\mathbb{G}_{4y}^{\alpha(1)} - \mathbb{G}_{4y}^{\beta(1)} + \mathbb{G}_{xz}^{(1)} + 2\mathbb{G}_{xz}^{(2)} \\
G_{yyyy} &= \mathbb{G}_0^{(1)} + 2\mathbb{G}_0^{(2)} - \mathbb{G}_{4u}^{(1)} + \mathbb{G}_{4v}^{(1)} + 2\mathbb{G}_4^{(1)} - 2\mathbb{G}_u^{(1)} - 4\mathbb{G}_u^{(2)} - 2\mathbb{G}_v^{(1)} - 4\mathbb{G}_v^{(2)} \\
G_{yyzz} &= \mathbb{G}_0^{(1)} - \mathbb{G}_{4u}^{(1)} - \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(1)} - \mathbb{G}_v^{(1)} \\
G_{yyzx} &= 2\mathbb{G}_{4y}^{\beta(1)} + \mathbb{G}_{xz}^{(1)} \\
G_{zzzz} &= \mathbb{G}_0^{(1)} + 2\mathbb{G}_0^{(2)} + 2\mathbb{G}_{4u}^{(1)} + 2\mathbb{G}_4^{(1)} + 4\mathbb{G}_u^{(1)} + 8\mathbb{G}_u^{(2)} \\
G_{zzzx} &= \mathbb{G}_{4y}^{\alpha(1)} - \mathbb{G}_{4y}^{\beta(1)} + \mathbb{G}_{xz}^{(1)} + 2\mathbb{G}_{xz}^{(2)} \\
G_{yzyz} &= \mathbb{G}_0^{(2)} - \mathbb{G}_{4u}^{(1)} - \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(2)} - \mathbb{G}_v^{(2)} \\
G_{yzxy} &= 2\mathbb{G}_{4y}^{\beta(1)} + \mathbb{G}_{xz}^{(2)} \\
G_{zxzx} &= \mathbb{G}_0^{(2)} - \mathbb{G}_{4u}^{(1)} + \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(2)} + \mathbb{G}_v^{(2)} \\
G_{xyxy} &= \mathbb{G}_0^{(2)} + 2\mathbb{G}_{4u}^{(1)} - \mathbb{G}_4^{(1)} - 2\mathbb{G}_u^{(2)}
\end{aligned}
$$

* Rank 4 tensor (ssa).

$$
\begin{bmatrix}
0 & G_{xxyy} & G_{xxzz} & 0 & G_{xxzx} & 0 \\
-G_{xxyy} & 0 & G_{yyzz} & 0 & G_{yyzx} & 0 \\
-G_{xxzz} & -G_{yyzz} & 0 & 0 & G_{zzzx} & 0 \\
0 & 0 & 0 & 0 & 0 & G_{yzxy} \\
-G_{xxzx} & -G_{yyzx} & -G_{zzzx} & 0 & 0 & 0 \\
0 & 0 & 0 & -G_{yzxy} & 0 & 0
\end{bmatrix}
$$

$$
\begin{aligned}
G_{xxyy} &= -2\mathbb{G}_{v}^{(3)} + 4\mathbb{Q}_{xyz}^{(1)} \\
G_{xxzz} &= 3\mathbb{G}_{u}^{(3)} - \mathbb{G}_{v}^{(3)} - 4\mathbb{Q}_{xyz}^{(1)} \\
G_{xxzx} &= \mathbb{G}_{xz}^{(3)} - 2\mathbb{Q}_{y}^{(1)} + 2\mathbb{Q}_{y}^{\alpha(1)} + 2\mathbb{Q}_{y}^{\beta(1)} \\
G_{yyzz} &= 3\mathbb{G}_{u}^{(3)} + \mathbb{G}_{v}^{(3)} + 4\mathbb{Q}_{xyz}^{(1)} \\
G_{yyzx} &= \mathbb{G}_{xz}^{(3)} - 4\mathbb{Q}_{y}^{\beta(1)} \\
G_{zzzx} &= \mathbb{G}_{xz}^{(3)} + 2\mathbb{Q}_{y}^{(1)} - 2\mathbb{Q}_{y}^{\alpha(1)} + 2\mathbb{Q}_{y}^{\beta(1)} \\
G_{yzxy} &= \mathbb{Q}_{y}^{(1)} + 4\mathbb{Q}_{y}^{\alpha(1)}
\end{aligned}
$$

* Rank 4 tensor (aas).

$$
\begin{bmatrix}
G_{yzyz} & 0 & G_{yzxy} \\
0 & G_{zxzx} & 0 \\
G_{yzxy} & 0 & G_{xyxy}
\end{bmatrix}
$$

$$
\begin{aligned}
G_{yzyz} &= \mathbb{G}_{0}^{(3)} - 2\mathbb{G}_{u}^{(4)} + 2\mathbb{G}_{v}^{(4)} \\
G_{yzxy} &= 2\mathbb{G}_{xz}^{(4)} \\
G_{zxzx} &= \mathbb{G}_{0}^{(3)} - 2\mathbb{G}_{u}^{(4)} - 2\mathbb{G}_{v}^{(4)} \\
G_{xyxy} &= \mathbb{G}_{0}^{(3)} + 4\mathbb{G}_{u}^{(4)}
\end{aligned}
$$

* Rank 4 tensor (aaa).

$$
\begin{bmatrix}
0 & 0 & G_{yzxy} \\
0 & 0 & 0 \\
-G_{yzxy} & 0 & 0
\end{bmatrix}
$$

$$
G_{yzxy} = \mathbb{Q}_{y}^{(2)}
$$

* Rank 4 tensor (sa).

$$
\begin{bmatrix}
0 & G_{xxzx} & 0 \\
0 & G_{yyzx} & 0 \\
0 & G_{zzzx} & 0 \\
G_{yzyz} & 0 & G_{yzxy} \\
0 & G_{zxzx} & 0 \\
G_{xyyz} & 0 & G_{xyxy}
\end{bmatrix}
$$

$$
\begin{aligned}
G_{xxzx} &= 2\mathbb{G}_{xz}^{(5)} + \mathbb{Q}_{y}^{(3)} - \mathbb{Q}_{y}^{\alpha(2)} - \mathbb{Q}_{y}^{\beta(2)} \\
G_{yyzx} &= \mathbb{Q}_{y}^{(3)} + 2\mathbb{Q}_{y}^{(4)} + 2\mathbb{Q}_{y}^{\alpha(2)} \\
G_{zzzx} &= -2\mathbb{G}_{xz}^{(5)} + \mathbb{Q}_{y}^{(3)} - \mathbb{Q}_{y}^{\alpha(2)} + \mathbb{Q}_{y}^{\beta(2)} \\
G_{yzyz} &= -3\mathbb{G}_{u}^{(5)} - \mathbb{G}_{v}^{(5)} + \mathbb{Q}_{xyz}^{(2)} \\
G_{yzxy} &= \mathbb{G}_{xz}^{(5)} + \mathbb{Q}_{y}^{(4)} - \mathbb{Q}_{y}^{\alpha(2)} + \mathbb{Q}_{y}^{\beta(2)} \\
G_{zxzx} &= 3\mathbb{G}_{u}^{(5)} - \mathbb{G}_{v}^{(5)} + \mathbb{Q}_{xyz}^{(2)} \\
G_{xyyz} &= -\mathbb{G}_{xz}^{(5)} + \mathbb{Q}_{y}^{(4)} - \mathbb{Q}_{y}^{\alpha(2)} - \mathbb{Q}_{y}^{\beta(2)} \\
G_{xyxy} &= 2\mathbb{G}_{v}^{(5)} + \mathbb{Q}_{xyz}^{(2)}
\end{aligned}
$$

* Rank 4 tensor (as).

$$
\begin{bmatrix}
0 & 0 & 0 & G_{yzyz} & 0 & G_{yzxy} \\
G_{zxxx} & G_{zxyy} & G_{zxzz} & 0 & G_{zxzx} & 0 \\
0 & 0 & 0 & G_{xyyz} & 0 & G_{xyxy}
\end{bmatrix}
$$

$$
\begin{aligned}
G_{yzyz} &= -3\mathbb{G}_u^{(6)} - \mathbb{G}_v^{(6)} + \mathbb{Q}_{xyz}^{(3)} \\
G_{yzxy} &= -\mathbb{G}_{xz}^{(6)} + \mathbb{Q}_y^{(6)} - \mathbb{Q}_y^{\alpha(3)} - \mathbb{Q}_y^{\beta(3)} \\
G_{zxxx} &= 2\mathbb{G}_{xz}^{(6)} + \mathbb{Q}_y^{(5)} - \mathbb{Q}_y^{\alpha(3)} - \mathbb{Q}_y^{\beta(3)} \\
G_{zxyy} &= \mathbb{Q}_y^{(5)} + 2\mathbb{Q}_y^{(6)} + 2\mathbb{Q}_y^{\alpha(3)} \\
G_{zxzz} &= -2\mathbb{G}_{xz}^{(6)} + \mathbb{Q}_y^{(5)} - \mathbb{Q}_y^{\alpha(3)} + \mathbb{Q}_y^{\beta(3)} \\
G_{zxzx} &= 3\mathbb{G}_u^{(6)} - \mathbb{G}_v^{(6)} + \mathbb{Q}_{xyz}^{(3)} \\
G_{xyyz} &= \mathbb{G}_{xz}^{(6)} + \mathbb{Q}_y^{(6)} - \mathbb{Q}_y^{\alpha(3)} + \mathbb{Q}_y^{\beta(3)} \\
G_{xyxy} &= 2\mathbb{G}_v^{(6)} + \mathbb{Q}_{xyz}^{(3)}
\end{aligned}
$$

* Rank 4 tensor (s).

$$
\begin{bmatrix}
G_{xxxx} & G_{xxyy} & G_{xxzz} & 0 & G_{xxzx} & 0 & 0 & G_{xxxz} & 0 \\
G_{yyxx} & G_{yyyy} & G_{yyzz} & 0 & G_{yyzx} & 0 & 0 & G_{yyxz} & 0 \\
G_{zzxx} & G_{zzyy} & G_{zzzz} & 0 & G_{zzzx} & 0 & 0 & G_{zzxz} & 0 \\
0 & 0 & 0 & G_{yzyz} & 0 & G_{yzxy} & G_{yzzy} & 0 & G_{yzyx} \\
G_{zxxx} & G_{zxyy} & G_{zxzz} & 0 & G_{zxzx} & 0 & 0 & G_{zxxz} & 0 \\
0 & 0 & 0 & G_{xyyz} & 0 & G_{xyxy} & G_{xyzy} & 0 & G_{xyyx}
\end{bmatrix}
$$

$$G_{xxxx} = \mathbb{G}_0^{(1)} + 2\mathbb{G}_0^{(2)} - \mathbb{G}_{4u}^{(1)} - \mathbb{G}_{4v}^{(1)} + 2\mathbb{G}_4^{(1)} - 2\mathbb{G}_u^{(1)} - 4\mathbb{G}_u^{(2)} + 2\mathbb{G}_v^{(1)} + 4\mathbb{G}_v^{(2)}$$

$$G_{xxyy} = \mathbb{G}_0^{(1)} + 2\mathbb{G}_{4u}^{(1)} - \mathbb{G}_4^{(1)} - 2\mathbb{G}_u^{(1)} - 2\mathbb{G}_v^{(3)} + 4\mathbb{Q}_{xyz}^{(1)}$$

$$G_{xxzz} = \mathbb{G}_0^{(1)} - \mathbb{G}_{4u}^{(1)} + \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(1)} + 3\mathbb{G}_u^{(3)} + \mathbb{G}_v^{(1)} - \mathbb{G}_v^{(3)} - 4\mathbb{Q}_{xyz}^{(1)}$$

$$G_{xxzx} = -\mathbb{G}_{4y}^{\alpha(1)} - \mathbb{G}_{4y}^{\beta(1)} + \mathbb{G}_{xz}^{(1)} + 2\mathbb{G}_{xz}^{(2)} + \mathbb{G}_{xz}^{(3)} + 2\mathbb{G}_{xz}^{(5)} - 2\mathbb{Q}_y^{(1)} + \mathbb{Q}_y^{(3)} + 2\mathbb{Q}_y^{\alpha(1)} - \mathbb{Q}_y^{\alpha(2)} + 2\mathbb{Q}_y^{\beta(1)} - \mathbb{Q}_y^{\beta(2)}$$

$$G_{xxxz} = -\mathbb{G}_{4y}^{\alpha(1)} - \mathbb{G}_{4y}^{\beta(1)} + \mathbb{G}_{xz}^{(1)} + 2\mathbb{G}_{xz}^{(2)} + \mathbb{G}_{xz}^{(3)} - 2\mathbb{G}_{xz}^{(5)} - 2\mathbb{Q}_y^{(1)} - \mathbb{Q}_y^{(3)} + 2\mathbb{Q}_y^{\alpha(1)} + \mathbb{Q}_y^{\alpha(2)} + 2\mathbb{Q}_y^{\beta(1)} + \mathbb{Q}_y^{\beta(2)}$$

$$G_{yyxx} = \mathbb{G}_0^{(1)} + 2\mathbb{G}_{4u}^{(1)} - \mathbb{G}_4^{(1)} - 2\mathbb{G}_u^{(1)} + 2\mathbb{G}_v^{(3)} - 4\mathbb{Q}_{xyz}^{(1)}$$

$$G_{yyyy} = \mathbb{G}_0^{(1)} + 2\mathbb{G}_0^{(2)} - \mathbb{G}_{4u}^{(1)} + \mathbb{G}_{4v}^{(1)} + 2\mathbb{G}_4^{(1)} - 2\mathbb{G}_u^{(1)} - 4\mathbb{G}_u^{(2)} - 2\mathbb{G}_v^{(1)} - 4\mathbb{G}_v^{(2)}$$

$$G_{yyzz} = \mathbb{G}_0^{(1)} - \mathbb{G}_{4u}^{(1)} - \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(1)} + 3\mathbb{G}_u^{(3)} - \mathbb{G}_v^{(1)} + \mathbb{G}_v^{(3)} + 4\mathbb{Q}_{xyz}^{(1)}$$

$$G_{yyzx} = 2\mathbb{G}_{4y}^{\beta(1)} + \mathbb{G}_{xz}^{(1)} + \mathbb{G}_{xz}^{(3)} + \mathbb{Q}_y^{(3)} + 2\mathbb{Q}_y^{(4)} + 2\mathbb{Q}_y^{\alpha(2)} - 4\mathbb{Q}_y^{\beta(1)}$$

$$G_{yyxz} = 2\mathbb{G}_{4y}^{\beta(1)} + \mathbb{G}_{xz}^{(1)} + \mathbb{G}_{xz}^{(3)} - \mathbb{Q}_y^{(3)} - 2\mathbb{Q}_y^{(4)} - 2\mathbb{Q}_y^{\alpha(2)} - 4\mathbb{Q}_y^{\beta(1)}$$

$$G_{zzxx} = \mathbb{G}_0^{(1)} - \mathbb{G}_{4u}^{(1)} + \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(1)} - 3\mathbb{G}_u^{(3)} + \mathbb{G}_v^{(1)} + \mathbb{G}_v^{(3)} + 4\mathbb{Q}_{xyz}^{(1)}$$

$$G_{zzyy} = \mathbb{G}_0^{(1)} - \mathbb{G}_{4u}^{(1)} - \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(1)} - 3\mathbb{G}_u^{(3)} - \mathbb{G}_v^{(1)} - \mathbb{G}_v^{(3)} - 4\mathbb{Q}_{xyz}^{(1)}$$

$$G_{zzzz} = \mathbb{G}_0^{(1)} + 2\mathbb{G}_0^{(2)} + 2\mathbb{G}_{4u}^{(1)} + 2\mathbb{G}_4^{(1)} + 4\mathbb{G}_u^{(1)} + 8\mathbb{G}_u^{(2)}$$

$$G_{zzzx} = \mathbb{G}_{4y}^{\alpha(1)} - \mathbb{G}_{4y}^{\beta(1)} + \mathbb{G}_{xz}^{(1)} + 2\mathbb{G}_{xz}^{(2)} + \mathbb{G}_{xz}^{(3)} - 2\mathbb{G}_{xz}^{(5)} + 2\mathbb{Q}_y^{(1)} + \mathbb{Q}_y^{(3)} - 2\mathbb{Q}_y^{\alpha(1)} - \mathbb{Q}_y^{\alpha(2)} + 2\mathbb{Q}_y^{\beta(1)} + \mathbb{Q}_y^{\beta(2)}$$

$$G_{zzxz} = \mathbb{G}_{4y}^{\alpha(1)} - \mathbb{G}_{4y}^{\beta(1)} + \mathbb{G}_{xz}^{(1)} + 2\mathbb{G}_{xz}^{(2)} + \mathbb{G}_{xz}^{(3)} + 2\mathbb{G}_{xz}^{(5)} + 2\mathbb{Q}_y^{(1)} - \mathbb{Q}_y^{(3)} - 2\mathbb{Q}_y^{\alpha(1)} + \mathbb{Q}_y^{\alpha(2)} + 2\mathbb{Q}_y^{\beta(1)} - \mathbb{Q}_y^{\beta(2)}$$

$$G_{yzyz} = \mathbb{G}_0^{(2)} - \mathbb{G}_{4u}^{(1)} - \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(2)} - 3\mathbb{G}_u^{(5)} - \mathbb{G}_v^{(2)} - \mathbb{G}_v^{(5)} + \mathbb{Q}_{xyz}^{(2)}$$

$$G_{yzxy} = 2\mathbb{G}_{4y}^{\beta(1)} + \mathbb{G}_{xz}^{(2)} + \mathbb{G}_{xz}^{(5)} + \mathbb{Q}_y^{(1)} + \mathbb{Q}_y^{(4)} + 4\mathbb{Q}_y^{\alpha(1)} - \mathbb{Q}_y^{\alpha(2)} + \mathbb{Q}_y^{\beta(2)}$$

$$G_{yzzy} = \mathbb{G}_0^{(2)} - \mathbb{G}_{4u}^{(1)} - \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(2)} + 3\mathbb{G}_u^{(5)} - \mathbb{G}_v^{(2)} + \mathbb{G}_v^{(5)} - \mathbb{Q}_{xyz}^{(2)}$$

$$G_{yzyx} = 2\mathbb{G}_{4y}^{\beta(1)} + \mathbb{G}_{xz}^{(2)} - \mathbb{G}_{xz}^{(5)} + \mathbb{Q}_y^{(1)} - \mathbb{Q}_y^{(4)} + 4\mathbb{Q}_y^{\alpha(1)} + \mathbb{Q}_y^{\alpha(2)} - \mathbb{Q}_y^{\beta(2)}$$

$$G_{zxxx} = -\mathbb{G}_{4y}^{\alpha(1)} - \mathbb{G}_{4y}^{\beta(1)} + \mathbb{G}_{xz}^{(1)} + 2\mathbb{G}_{xz}^{(2)} - \mathbb{G}_{xz}^{(3)} + 2\mathbb{Q}_y^{(1)} - 2\mathbb{Q}_y^{\alpha(1)} - 2\mathbb{Q}_y^{\beta(1)}$$

$$G_{zxyy} = 2\mathbb{G}_{4y}^{\beta(1)} + \mathbb{G}_{xz}^{(1)} - \mathbb{G}_{xz}^{(3)} + 4\mathbb{Q}_y^{\beta(1)}$$

$$G_{zxzz} = \mathbb{G}_{4y}^{\alpha(1)} - \mathbb{G}_{4y}^{\beta(1)} + \mathbb{G}_{xz}^{(1)} + 2\mathbb{G}_{xz}^{(2)} - \mathbb{G}_{xz}^{(3)} - 2\mathbb{Q}_y^{(1)} + 2\mathbb{Q}_y^{\alpha(1)} - 2\mathbb{Q}_y^{\beta(1)}$$

$$G_{zxzx} = \mathbb{G}_0^{(2)} - \mathbb{G}_{4u}^{(1)} + \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(2)} + 3\mathbb{G}_u^{(5)} + \mathbb{G}_v^{(2)} - \mathbb{G}_v^{(5)} + \mathbb{Q}_{xyz}^{(2)}$$

$$G_{zxxz} = \mathbb{G}_0^{(2)} - \mathbb{G}_{4u}^{(1)} + \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(2)} - 3\mathbb{G}_u^{(5)} + \mathbb{G}_v^{(2)} + \mathbb{G}_v^{(5)} - \mathbb{Q}_{xyz}^{(2)}$$

$$G_{xyyz} = 2\mathbb{G}_{4y}^{\beta(1)} + \mathbb{G}_{xz}^{(2)} - \mathbb{G}_{xz}^{(5)} - \mathbb{Q}_y^{(1)} + \mathbb{Q}_y^{(4)} - 4\mathbb{Q}_y^{\alpha(1)} - \mathbb{Q}_y^{\alpha(2)} - \mathbb{Q}_y^{\beta(2)}$$

$$G_{xyxy} = \mathbb{G}_0^{(2)} + 2\mathbb{G}_{4u}^{(1)} - \mathbb{G}_4^{(1)} - 2\mathbb{G}_u^{(2)} + 2\mathbb{G}_v^{(5)} + \mathbb{Q}_{xyz}^{(2)}$$

$$G_{xyzy} = 2\mathbb{G}_{4y}^{\beta(1)} + \mathbb{G}_{xz}^{(2)} + \mathbb{G}_{xz}^{(5)} - \mathbb{Q}_y^{(1)} - \mathbb{Q}_y^{(4)} - 4\mathbb{Q}_y^{\alpha(1)} + \mathbb{Q}_y^{\alpha(2)} + \mathbb{Q}_y^{\beta(2)}$$

$$G_{xyyx} = \mathbb{G}_0^{(2)} + 2\mathbb{G}_{4u}^{(1)} - \mathbb{G}_4^{(1)} - 2\mathbb{G}_u^{(2)} - 2\mathbb{G}_v^{(5)} - \mathbb{Q}_{xyz}^{(2)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & G_{yzyz} & 0 & G_{yzxy} & G_{yzzy} & 0 & G_{yzyx} \\ G_{zxxx} & G_{zxyy} & G_{zxzz} & 0 & G_{zxzx} & 0 & 0 & G_{zxxz} & 0 \\ 0 & 0 & 0 & G_{xyyz} & 0 & G_{xyxy} & G_{xyzy} & 0 & G_{xyyx} \end{bmatrix}$$

$$G_{yzyz} = \mathbb{G}_0^{(3)} - 2\mathbb{G}_u^{(4)} - 3\mathbb{G}_u^{(6)} + 2\mathbb{G}_v^{(4)} - \mathbb{G}_v^{(6)} + \mathbb{Q}_{xyz}^{(3)}$$
$$G_{yzxy} = 2\mathbb{G}_{xz}^{(4)} - \mathbb{G}_{xz}^{(6)} + \mathbb{Q}_y^{(2)} + \mathbb{Q}_y^{(6)} - \mathbb{Q}_y^{\alpha(3)} - \mathbb{Q}_y^{\beta(3)}$$
$$G_{yzzy} = -\mathbb{G}_0^{(3)} + 2\mathbb{G}_u^{(4)} - 3\mathbb{G}_u^{(6)} - 2\mathbb{G}_v^{(4)} - \mathbb{G}_v^{(6)} + \mathbb{Q}_{xyz}^{(3)}$$
$$G_{yzyx} = -2\mathbb{G}_{xz}^{(4)} - \mathbb{G}_{xz}^{(6)} - \mathbb{Q}_y^{(2)} + \mathbb{Q}_y^{(6)} - \mathbb{Q}_y^{\alpha(3)} - \mathbb{Q}_y^{\beta(3)}$$
$$G_{zxxx} = 2\mathbb{G}_{xz}^{(6)} + \mathbb{Q}_y^{(5)} - \mathbb{Q}_y^{\alpha(3)} - \mathbb{Q}_y^{\beta(3)}$$
$$G_{zxyy} = \mathbb{Q}_y^{(5)} + 2\mathbb{Q}_y^{(6)} + 2\mathbb{Q}_y^{\alpha(3)}$$
$$G_{zxzz} = -2\mathbb{G}_{xz}^{(6)} + \mathbb{Q}_y^{(5)} - \mathbb{Q}_y^{\alpha(3)} + \mathbb{Q}_y^{\beta(3)}$$
$$G_{zxzx} = \mathbb{G}_0^{(3)} - 2\mathbb{G}_u^{(4)} + 3\mathbb{G}_u^{(6)} - 2\mathbb{G}_v^{(4)} - \mathbb{G}_v^{(6)} + \mathbb{Q}_{xyz}^{(3)}$$
$$G_{zxxz} = -\mathbb{G}_0^{(3)} + 2\mathbb{G}_u^{(4)} + 3\mathbb{G}_u^{(6)} + 2\mathbb{G}_v^{(4)} - \mathbb{G}_v^{(6)} + \mathbb{Q}_{xyz}^{(3)}$$
$$G_{xyyz} = 2\mathbb{G}_{xz}^{(4)} + \mathbb{G}_{xz}^{(6)} - \mathbb{Q}_y^{(2)} + \mathbb{Q}_y^{(6)} - \mathbb{Q}_y^{\alpha(3)} + \mathbb{Q}_y^{\beta(3)}$$
$$G_{xyxy} = \mathbb{G}_0^{(3)} + 4\mathbb{G}_u^{(4)} + 2\mathbb{G}_v^{(6)} + \mathbb{Q}_{xyz}^{(3)}$$
$$G_{xyzy} = -2\mathbb{G}_{xz}^{(4)} + \mathbb{G}_{xz}^{(6)} + \mathbb{Q}_y^{(2)} + \mathbb{Q}_y^{(6)} - \mathbb{Q}_y^{\alpha(3)} + \mathbb{Q}_y^{\beta(3)}$$
$$G_{xyyx} = -\mathbb{G}_0^{(3)} - 4\mathbb{G}_u^{(4)} + 2\mathbb{G}_v^{(6)} + \mathbb{Q}_{xyz}^{(3)}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} G_{xxxx} & 0 & G_{xxxz} \\ 0 & G_{yyyy} & 0 \\ G_{zzzx} & 0 & G_{zzzz} \\ G_{yyzx} & 0 & G_{yyzz} \\ G_{zzxx} & 0 & G_{zzxz} \\ 0 & G_{xxyy} & 0 \\ 0 & G_{yzzy} & 0 \\ G_{zxxx} & 0 & G_{zxxz} \\ G_{xxyy} + G_{yyzz} - G_{yzzy} - G_{zxxz} + G_{zzxx} & 0 & G_{xyyz} \\ 0 & G_{xyyz} + \frac{G_{zzxx}}{2} - \frac{G_{zzzz}}{2} & 0 \end{bmatrix}$$

$$G_{xxxx} = 3\mathbb{G}_0^{(1)} - \mathbb{G}_{4u}^{(1)} - \mathbb{G}_{4v}^{(1)} + 2\mathbb{G}_4^{(1)} - 2\mathbb{G}_u^{(1)} - 4\mathbb{G}_u^{(2)} + 2\mathbb{G}_v^{(1)} + 4\mathbb{G}_v^{(2)}$$
$$G_{xxxz} = -\mathbb{G}_{4y}^{\alpha(1)} - \mathbb{G}_{4y}^{\beta(1)} + \mathbb{G}_{xz}^{(1)} + 2\mathbb{G}_{xz}^{(2)} + \mathbb{G}_{xz}^{(3)} - 2\mathbb{G}_{xz}^{(5)}$$
$$G_{yyyy} = 3\mathbb{G}_0^{(1)} - \mathbb{G}_{4u}^{(1)} + \mathbb{G}_{4v}^{(1)} + 2\mathbb{G}_4^{(1)} - 2\mathbb{G}_u^{(1)} - 4\mathbb{G}_u^{(2)} - 2\mathbb{G}_v^{(1)} - 4\mathbb{G}_v^{(2)}$$
$$G_{zzzx} = \mathbb{G}_{4y}^{\alpha(1)} - \mathbb{G}_{4y}^{\beta(1)} + \mathbb{G}_{xz}^{(1)} + 2\mathbb{G}_{xz}^{(2)} + \mathbb{G}_{xz}^{(3)} - 2\mathbb{G}_{xz}^{(5)}$$
$$G_{zzzz} = 3\mathbb{G}_0^{(1)} + 2\mathbb{G}_{4u}^{(1)} + 2\mathbb{G}_4^{(1)} + 4\mathbb{G}_u^{(1)} + 8\mathbb{G}_u^{(2)}$$
$$G_{yyzx} = 2\mathbb{G}_{4y}^{\beta(1)} + \mathbb{G}_{xz}^{(1)} + \mathbb{G}_{xz}^{(3)}$$
$$G_{yyzz} = \mathbb{G}_0^{(1)} - \mathbb{G}_{4u}^{(1)} - \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(1)} + 3\mathbb{G}_u^{(3)} - \mathbb{G}_v^{(1)} + \mathbb{G}_v^{(3)}$$
$$G_{zzxx} = \mathbb{G}_0^{(1)} - \mathbb{G}_{4u}^{(1)} + \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(1)} - 3\mathbb{G}_u^{(3)} + \mathbb{G}_v^{(1)} + \mathbb{G}_v^{(3)}$$
$$G_{zzxz} = \mathbb{G}_{4y}^{\alpha(1)} - \mathbb{G}_{4y}^{\beta(1)} + \mathbb{G}_{xz}^{(1)} + 2\mathbb{G}_{xz}^{(2)} + \mathbb{G}_{xz}^{(3)} + 2\mathbb{G}_{xz}^{(5)}$$
$$G_{xxyy} = \mathbb{G}_0^{(1)} + 2\mathbb{G}_{4u}^{(1)} - \mathbb{G}_4^{(1)} - 2\mathbb{G}_u^{(1)} - 2\mathbb{G}_v^{(3)}$$
$$G_{yzzy} = \mathbb{G}_0^{(1)} - \mathbb{G}_{4u}^{(1)} - \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(2)} + 3\mathbb{G}_u^{(5)} - \mathbb{G}_v^{(2)} + \mathbb{G}_v^{(5)}$$
$$G_{zxxx} = -\mathbb{G}_{4y}^{\alpha(1)} - \mathbb{G}_{4y}^{\beta(1)} + \mathbb{G}_{xz}^{(1)} + 2\mathbb{G}_{xz}^{(2)} - \mathbb{G}_{xz}^{(3)}$$
$$G_{zxxz} = \mathbb{G}_0^{(1)} - \mathbb{G}_{4u}^{(1)} + \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(2)} - 3\mathbb{G}_u^{(5)} + \mathbb{G}_v^{(2)} + \mathbb{G}_v^{(5)}$$
$$G_{xyyz} = 2\mathbb{G}_{4y}^{\beta(1)} + \mathbb{G}_{xz}^{(2)} - \mathbb{G}_{xz}^{(5)}$$